Tables of HAST-LNC_paper

Table 1: Dataset Attributes and Description

Attribute	Details
Dataset Name	Infant Cry Audio Corpus (Donate-a-Cry)
Source	https://www.kaggle.com/datasets/bhoomikavalani/donateacrycorpusfeaturesdataset
Data Collection Method	User-uploaded infant cry recordings
Audio File Format	WAV file
Total Audio Files	457
Audio Nature	Real-world, unmodified, unchecked recordings
Audio Duration	6sec - 7sec
Audio File Size Range	102KB - 110KB

Table 2 Classification Model Parameters and Suggested Values

Parameter	Range	Suggested Value
Audio Sampling Rate (Hz)	8,000 - 44,100	16,000
Cry Segment Duration (seconds)	1 - 5	2 - 3
Mel Frequency Bands	64 - 128	128
AST Embedding Dimension	256 - 768	768
Logistic Regression Regularization (C)	0.01 - 10	1.0
MLP Hidden Layers	1 - 3	2
MLP Hidden Neurons	64 - 512	128, 256
Learning Rate	0.0001 - 0.01	0.001
Batch Size	8 - 64	16, 32
Number of Training Epochs	20 - 100	50

Table 3: Original Dataset: Class-wise Distribution of Cry Samples

Cry Class Categories	Count
discomfort	27
hungry	382
tired	24
burping	8
belly pain	16
Total Cry Samples	457

Table 4: After Preprocessing and Feature Extraction both done in single code.

Metric	Value
Total usable samples (X, y)	457
Feature dimension	768
Unique classes	[2, 3, 4]

Table 5: Unique and valid cry samples

Dataset Part	Unique Labels
y	[2, 3, 4]
y_test	[2, 3, 4]

Table 6: Class Balancing

Class ID	Class Label	Class Weight
0	belly_pain	5.6154
1	burping	12.1667
2	discomfort	3.3182
3	hungry	0.2393
4	tired	3.8421

Table 7: Audio samples are into x and y labels.

Metric	Value
X shape	(457, 768)
y max label	4
Skipped files	0

Table 8: Label Vectorization

Index	Label Name
0	belly pain
1	burping
2	discomfort
3	hungry
4	tired

Table 9: Labels after training and testing.

Dataset	Number of Samples
Training samples	365
Testing samples	92

Table 10: Architecture of Logistic Regression

Layer	Type	Units / Parameters	Activation	Regularization
1	Input Layer	768 features (AST embedding)	-	Feature scaling applied
2	Linear Output Layer (Dense)	Number of classes (k)	Softmax	L2 regularization (optional)

Table 11: Labels after merging

Label ID	Class Name
2	discomfort
3	hungry

4	tired

Table 12: Finding Unique samples after merging

Variable	**Unique Values**
y unique	[2, 3, 4]
merged_y unique	[2, 3, 4]

Table 13: MLP Classifier Architecture

Layer	**Type**	**Units / Parameters**	**Activation**	**Regularization**
1	Input Layer	768 features (AST embedding)	-	Standard Scaling applied
2	Hidden Layer 1 (Dense)	256 neurons	ReLU	Dropout / L2 (if used)
3	Hidden Layer 2 (Dense)	128 neurons	ReLU	Dropout / L2 (if used)
4	Output Layer (Dense)	Number of classes (k)	Softmax	-

Table 14: Train–Test Data Split Configuration

Dataset Portion	**Percentage**	**Purpose**
Training Set	80%	Model learning
Testing Set	20%	Performance evaluation

Table 15: Overall Performance Metrics

Model	**Accuracy**	**Precision**	**Recall**	**F1 Score**	**MSE**	**MCC**
MLP Classifier	0.93	0.59	0.13	0.21	0.72	0.02
Logistic Regression	0.88	0.78	0.88	0.83	0.11	-

Table 16: Model Metrics Comparison

Metric	**Logistic Regression**	**MLP Classifier**
Overall Accuracy	**76.09%**	**82.61%**
Macro Avg Precision	0.27	0.28
Macro Avg Recall	0.25	0.33
Macro Avg F1 - score	0.25	0.30
Weighted Avg Precision	0.73	0.71
Weighted Avg Recall	0.76	0.83
Weighted Avg F1-score	0.74	0.76

Table 17: Comparison with the Previous Infant Cry Classification Studies

Aspect	**Previous Studies**	**Proposed Transformer Approach**
Feature Extraction	Uses handcrafted features like MFCC and spectrograms	Uses pretrained embeddings capturing patterns automatically
Model Complexity	Deep networks with many parameters	Lightweight classifiers with fewer parameters
Training Time	Longer due to training deep models	Faster since embeddings are pretrained
Computational Cost	High GPU usage required	Lower cost, suitable for real-time use
Handling Acoustic Variability	Needs feature engineering and augmentation	Embeddings generalize well to variations
Dataset Dependency	Often needs large or augmented datasets	Works well on moderately sized data
Interpretability	Deep models act as black boxes	Probabilities are easier to interpret
Performance Stability	Risk of overfitting on small datasets	Pretrained features reduce overfitting
Deployment Suitability	Hard to deploy on edge devices	Practical for monitoring and healthcare systems
Overall Contribution	Focus on complex architectures for accuracy	Shows simple models + good embeddings work well

Table 18: Performance Comparison with Previous Studies

Study / Approach	**Target Task**	**Accuracy**	**Key Remarks**
Traditional ML (SVM, GMM, k-NN based methods) [23]	Binary or multi-class cry classification	81 - 88%	Strong dependence on handcrafted features like MFCC
CNN / CRNN based models [23]	Multi-class infant cry detection	86 - 91%	Good feature learning but computationally heavy
Hybrid Deep Models (CNN+RNN / Attention networks) [23]	Multi-class cry classification	88 - 94%	Improved contextual modeling but prone to overfitting
Efficient Graph + AlgNet [23]	Multi-class infant cry classification	87 - 94%	Combines hybrid features with attention networks
H**AST-LNC** (**AST - MLP**)	Multi-class infant cry cause detection (hunger, discomfort, tiredness, etc.)	70 - 89%	Achieves strong performance with lower complexity and faster convergence

Figures of HAST-LNC_paper

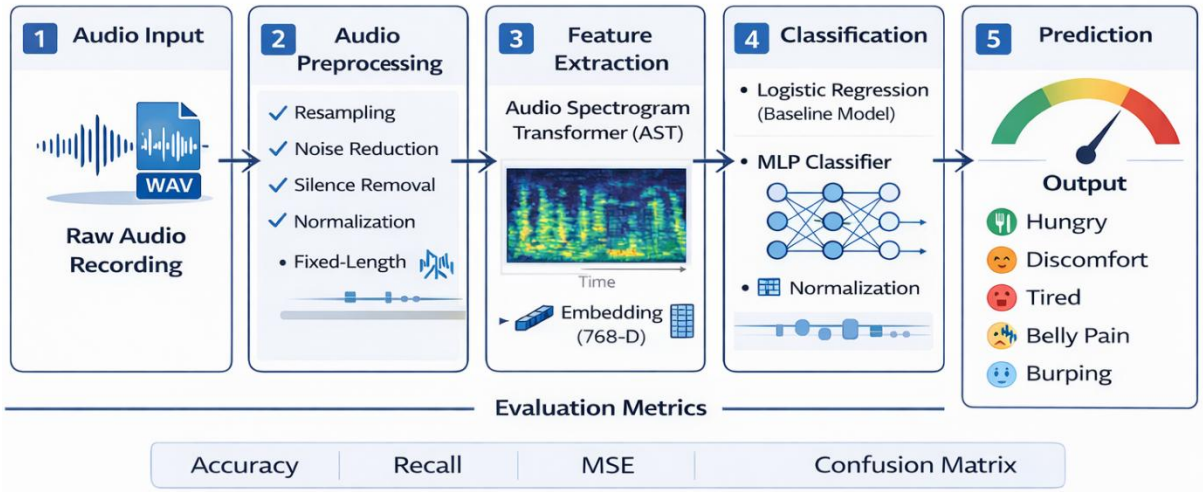


Fig 1. General cry type Multiclass Cry Classification Performance

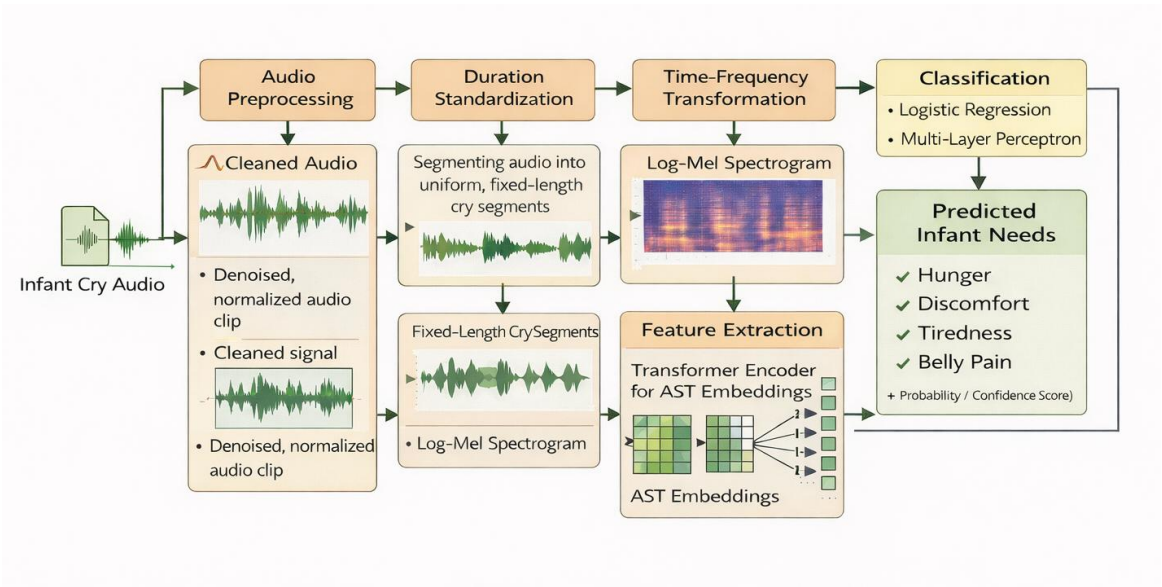


Fig. 2 Workflow diagram of proposed system

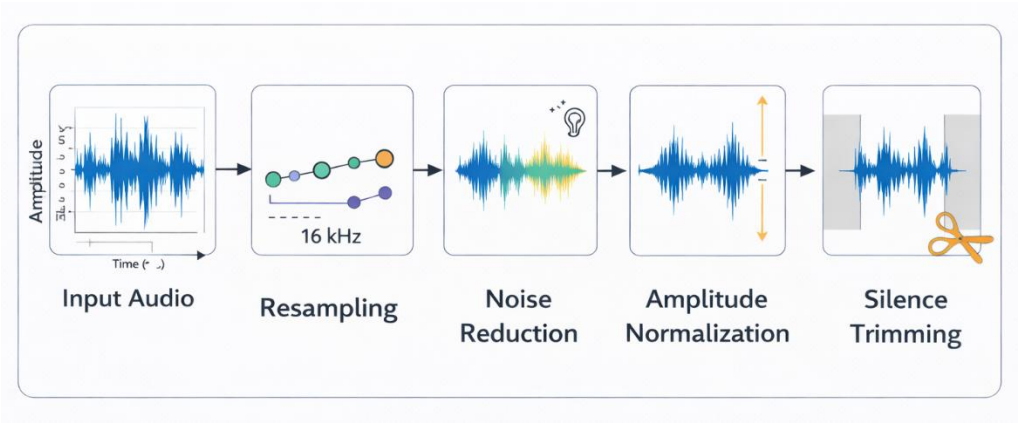
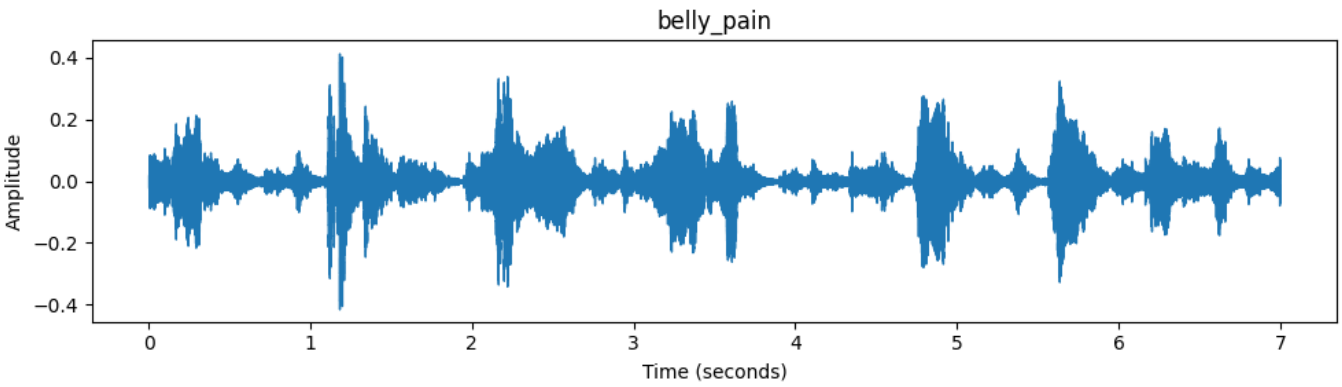
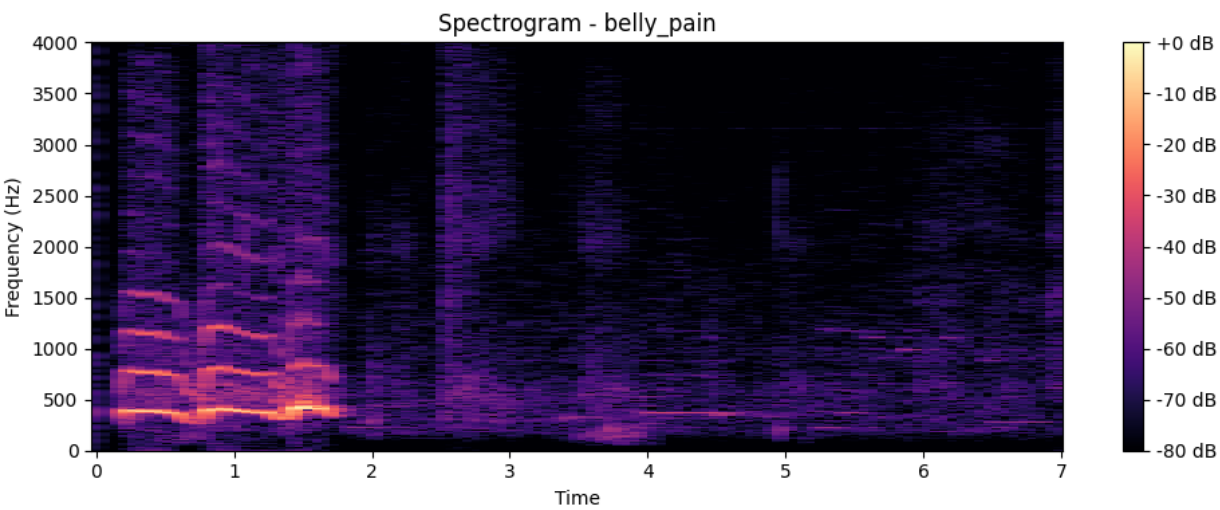


Fig 3: Audio Pre-processing flow

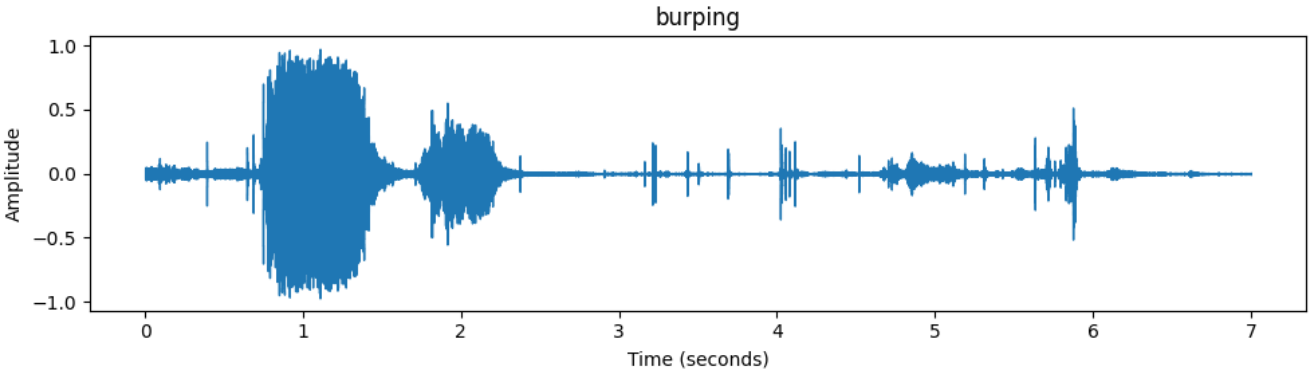


(a)

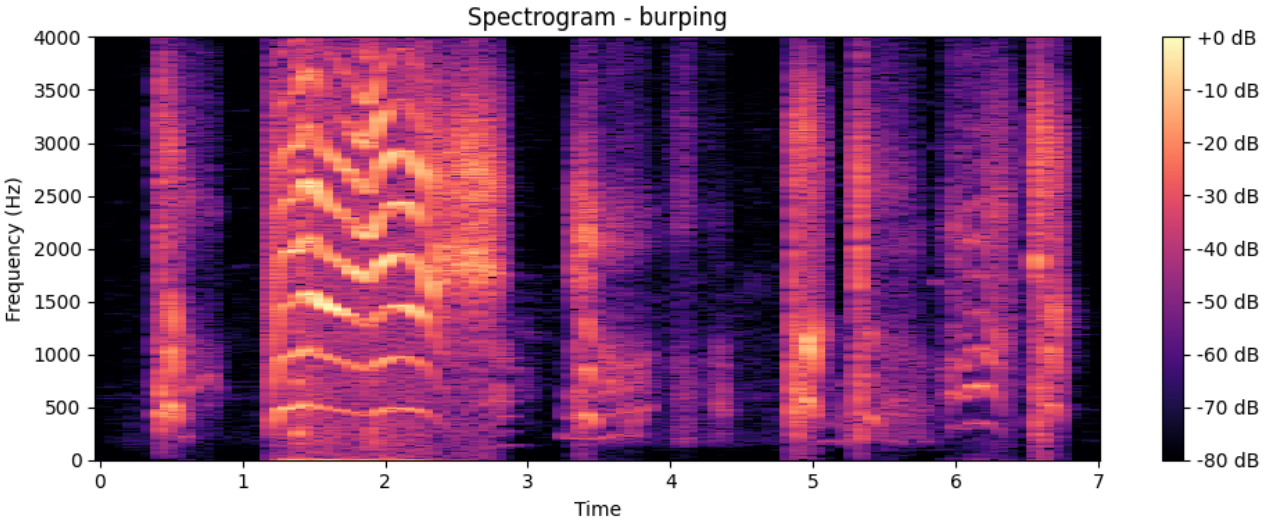


(b)

Fig 4: Belly pain infant cry: (a) the waveform representation, (b) the spectrogram.

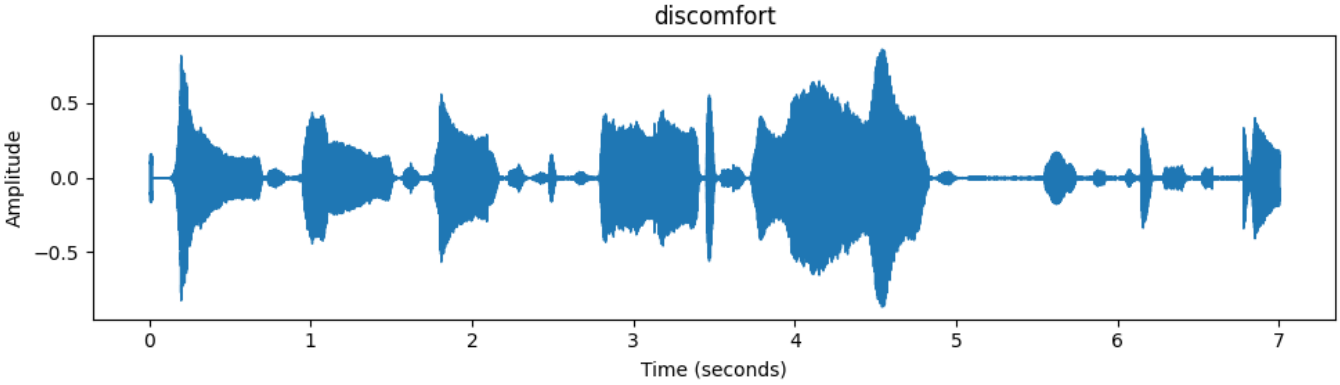


(a)

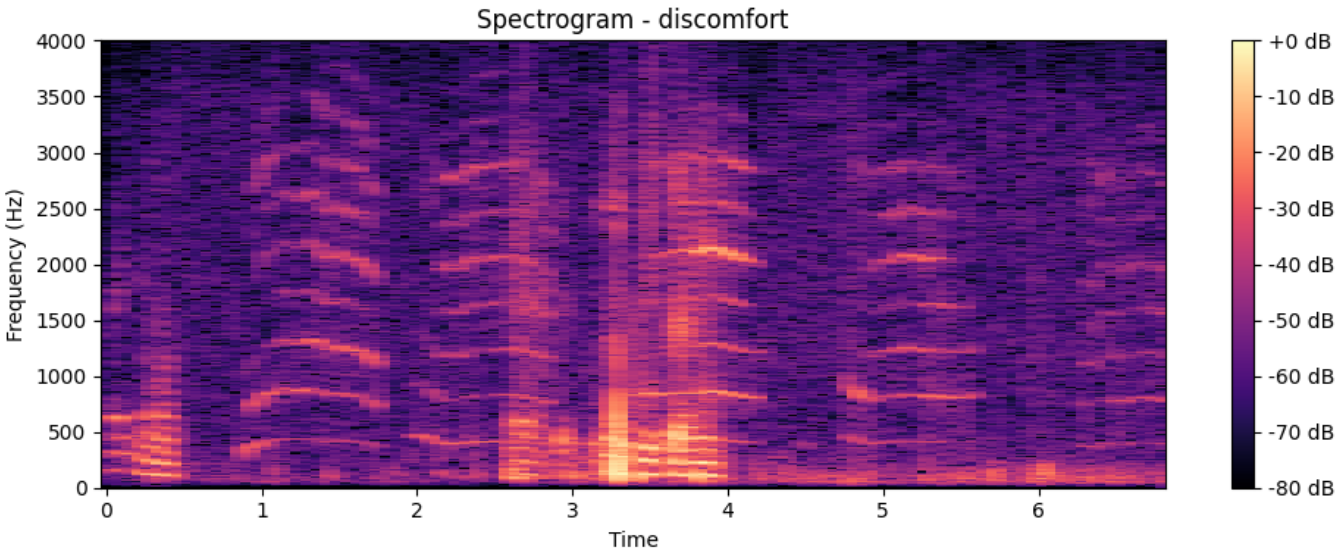


(b)

Fig 5: Burping infant cry: (a) the waveform representation, (b) the spectrogram.

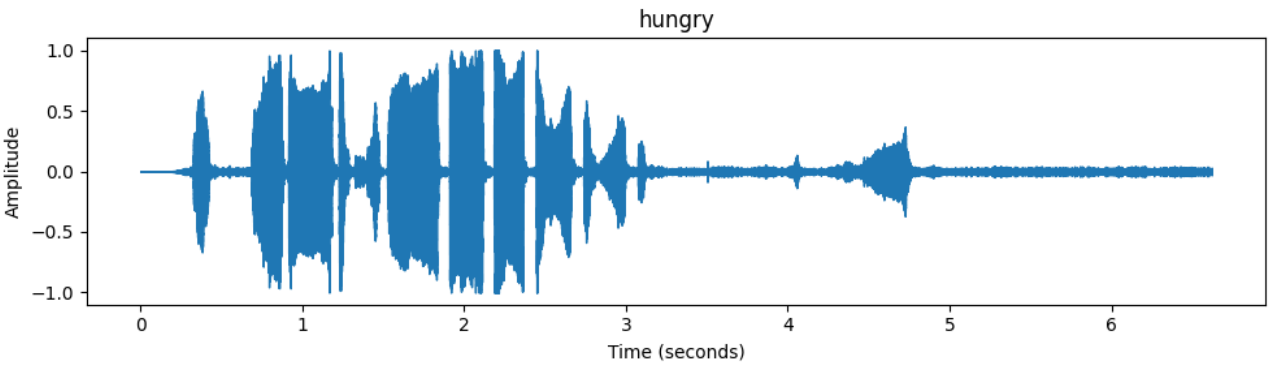


(a)



(b)

Fig 6: Discomfort infant cry: (a) the waveform representation, (b) the spectrogram.



(a)

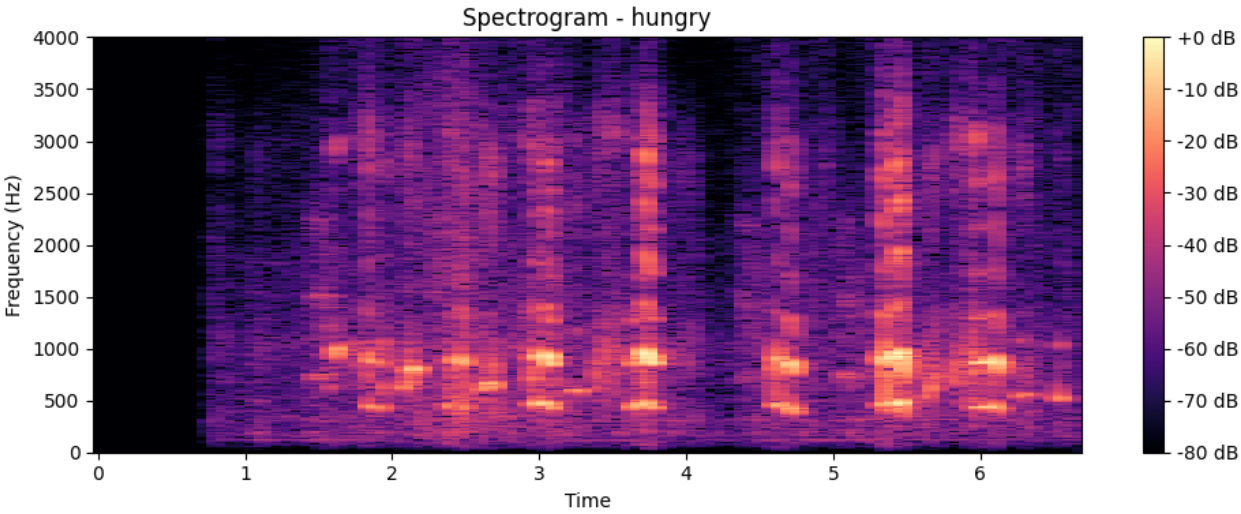


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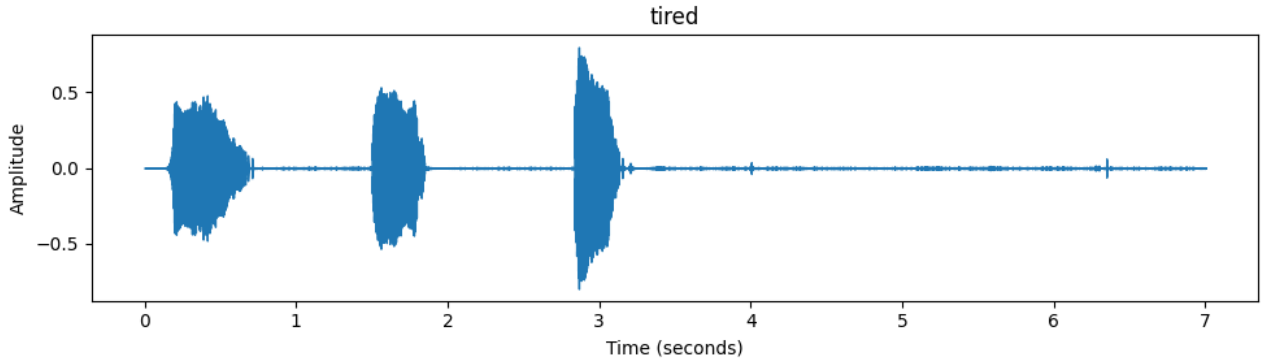
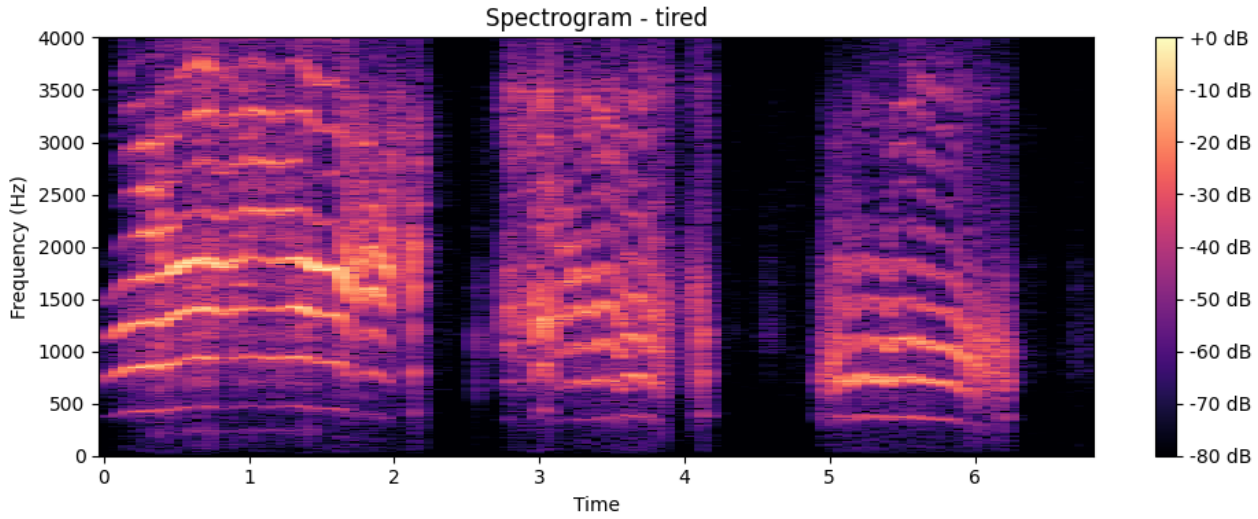
Fig 7: Hungry infant cry: (a) the waveform representation, (b) the spectrogram.



(a)



(b)

Fig 8: Tired infant cry: (a) the waveform representation, (b) the spectrogram.

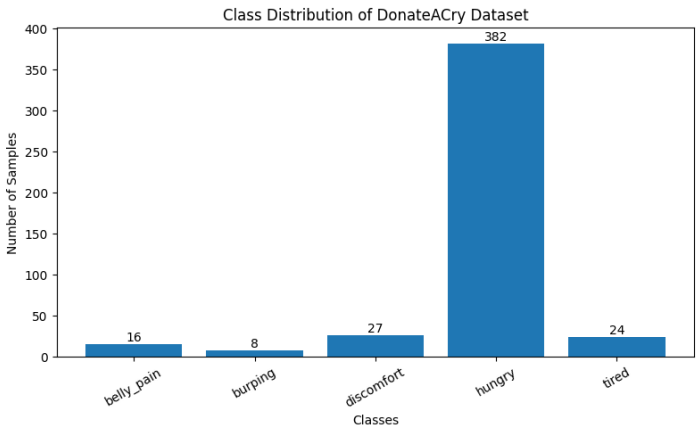


Fig 9: Class Distribution of dataset before balancing

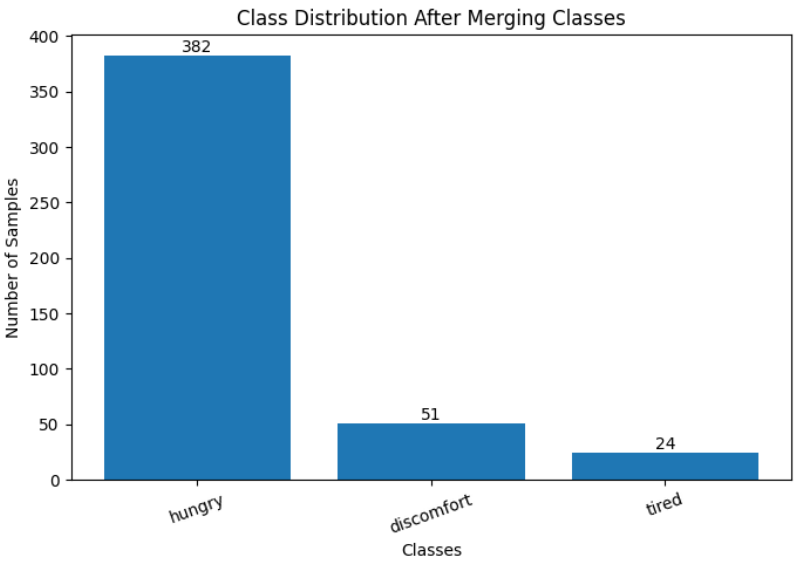


Fig 10: Class Distribution of dataset after balancing

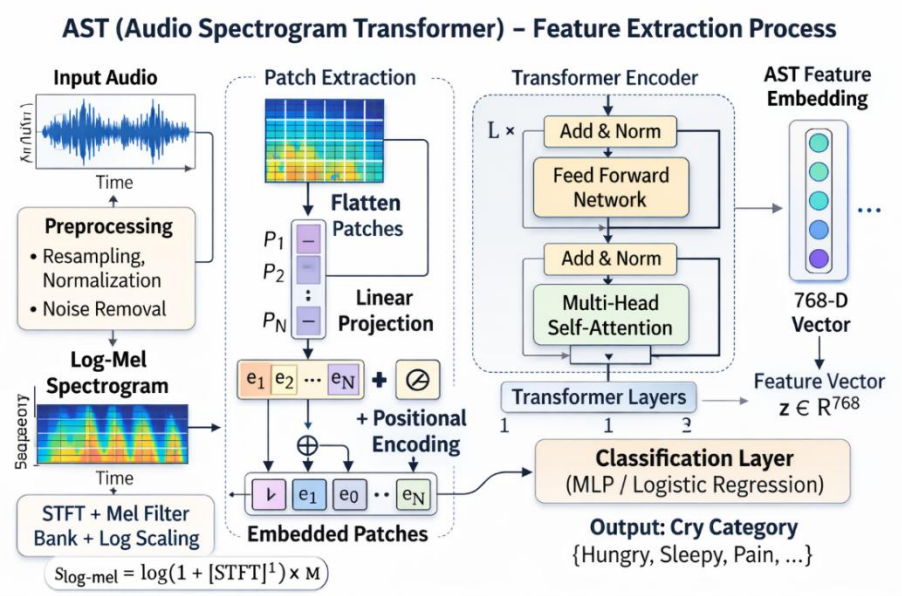


Fig 11: Working flow Audio Spectrogram Transfer for feature extraction process

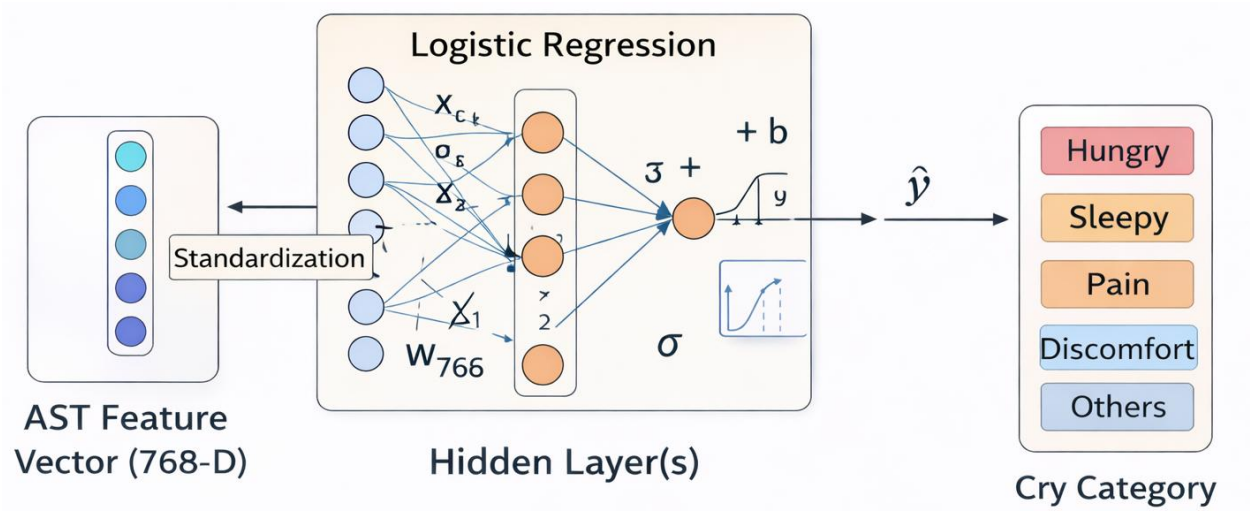


Fig 12: Logistic Regression Architecture

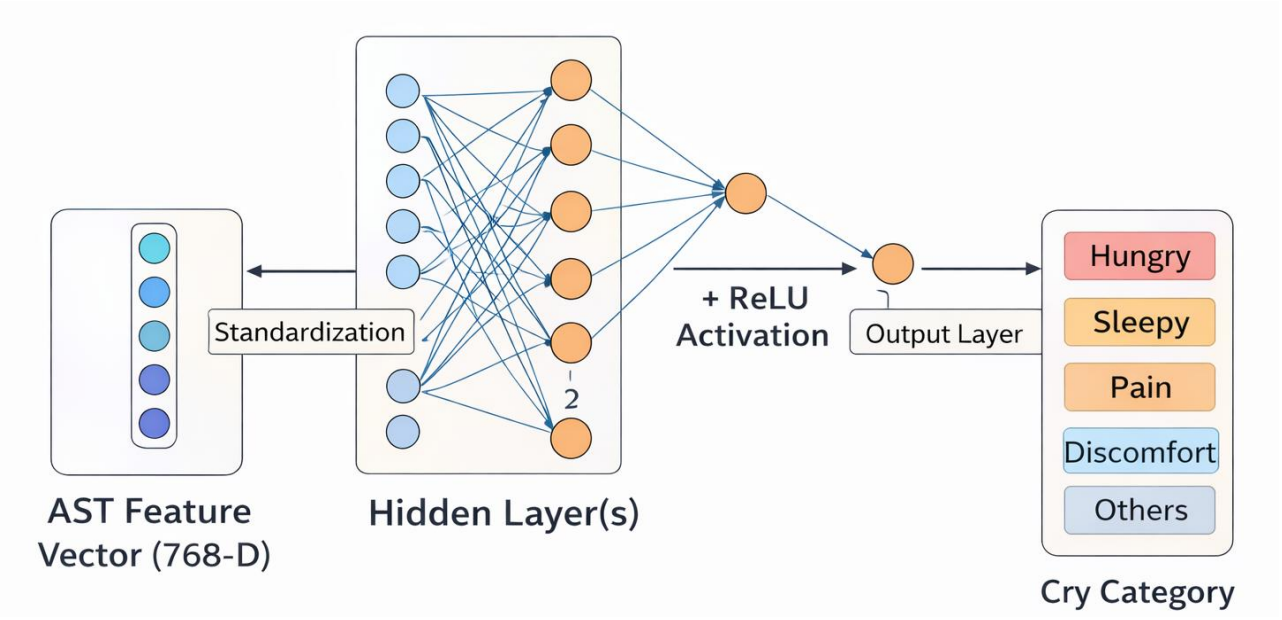


Fig 14: Working Design of MLP Classification

```
MLPClassifier
MLPClasstfier(hidden_layer_sizes=(256, 128), max_iter=500, random_state)
```

Fig 15: MLP Classifier

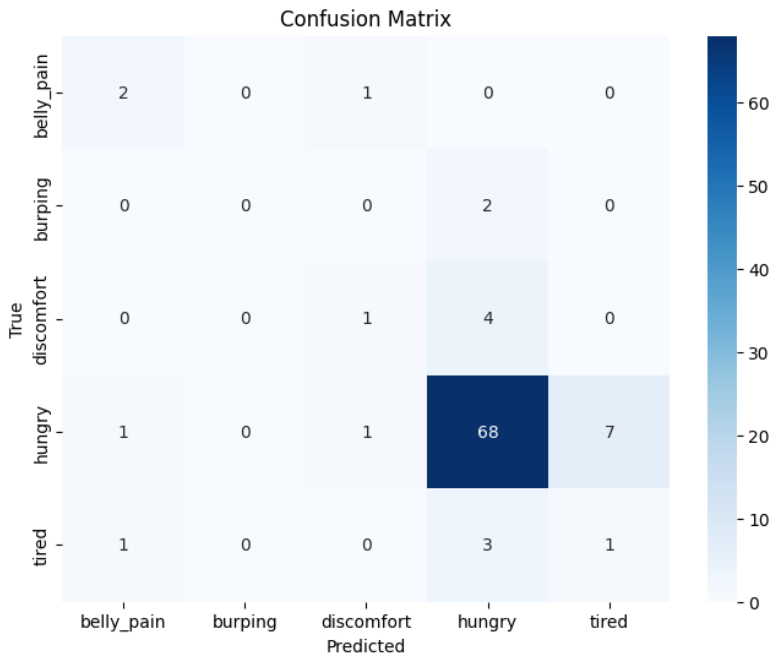


Fig 16: confusion Matrix

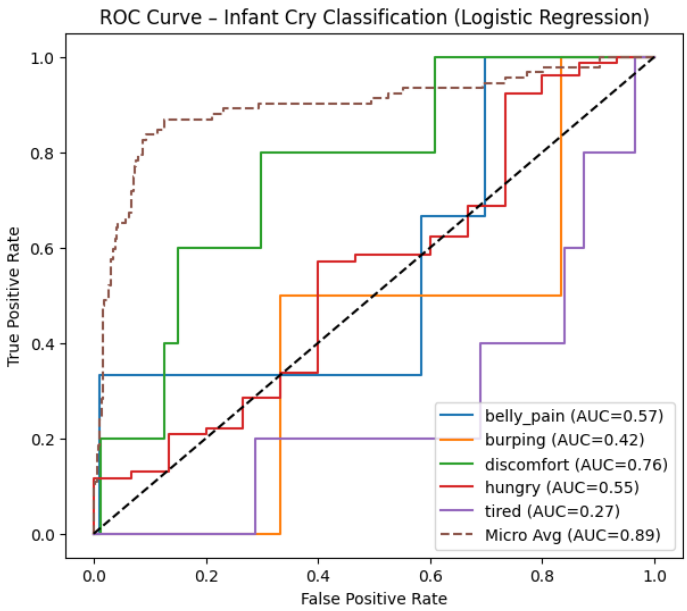


Fig 17: ROC curve of Logistic Regression

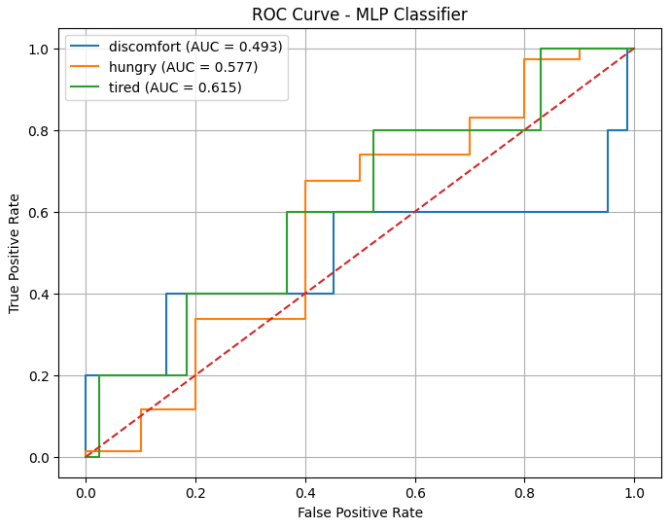


Fig 18: ROC curve of MLP CLassifier

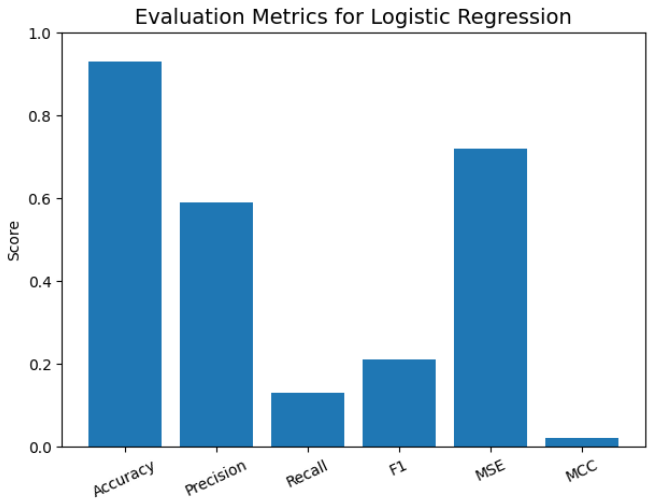


Fig 19: Evaluation metrics of Logistic Regression

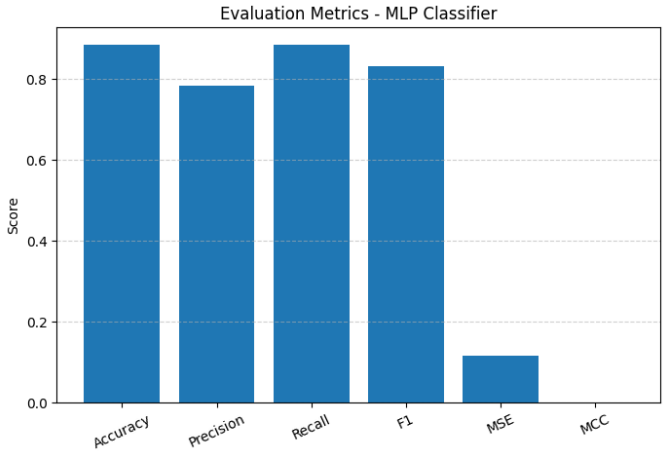


Fig 20: Evaluation metrics of MLP Classifier

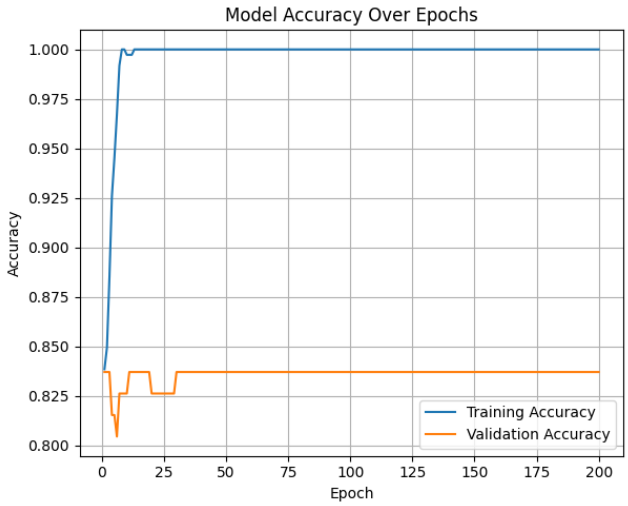


Fig 21: Accuracy Evaluation for MLP with Fusion

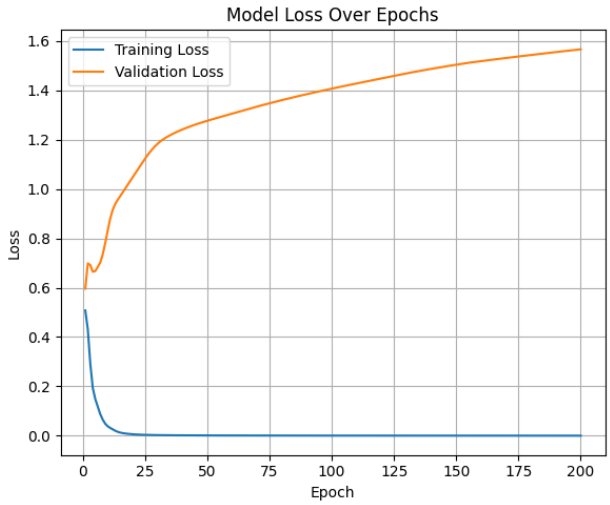


Fig 22: Loss Evaluation for MLP with Fusion